ClinVar exercise:

Disease	description	gene	Example variant	Mutation type	Functional effect
Thalassemia	Reduced functional hemoglobin production -> anemia	HBB	NM_000518.4(HBB):c.-122T>C	Single nucleotide variant	Partial LOF
Cystic fibrosis	Thick mucus in lung	CFTR	NM_000492.4(CFTR):c.3964-1G>C	Single nucleotide variant (splice acceptor variant)	LOF
Tay sachs disease	neurodegeneration	HEXA	NM_000520.6(HEXA):c.1422-1G>A	Single nucleotide variant (splice acceptor variant)	LOF

Ex 2 UniProt:

- Uniprot ID: Q12888
- Protein Name: TP53- binding protein 1
- Length:1972
- Keywords: (species: homo sapiens), DNA, double strand breaks, repair, telomere dynamics, class switching recombination (antibody genesis).
- Function: Plays a key role in the repair of double-strand DNA breaks (DSBs) in response to DNA damage by promoting non-homologous end joining (NHEJ)-mediated repair of DSBs and specifically counteracting the function of the homologous recombination (HR) repair protein BRCA1.

- What domains are there? The Tudor-like region mediates binding to histone H4 dimethylated at 'Lys-20' . The UDR (ubiquitin-dependent recruitment) motif specifically recognizes and binds histone H2A monoubiquitinated at 'Lys-15' (H2AK15ub)
- DNA-binding? Ligand-binding? DNA binding (histones)

- Gene Ontology?

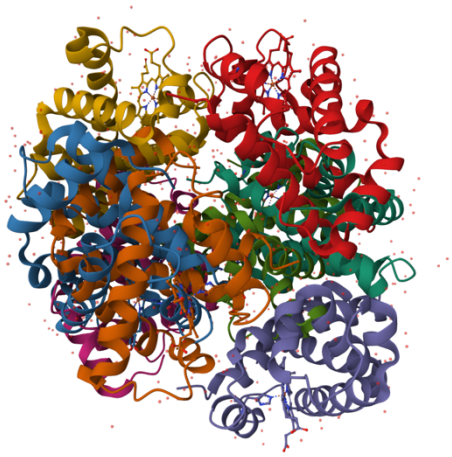
ASPECT	TERM
Molecular Function	damaged DNA binding Source:Ensembl
Molecular Function	histone binding Source:GO_Central
Molecular Function	histone H4K20me methyltransferase activity Source:Ensembl
Molecular Function	histone H4K20me2 reader activity Source:UniProtKB 4 Publications
Molecular Function	histone reader activity Source:UniProt 2 Publications
Molecular Function	p53 binding Source:AgBase 2 Publications
Molecular	RNA polymerase II-specific DNA-binding transcription factor binding

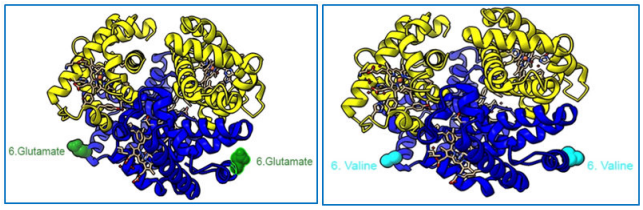
- Which mutations & diseases? A chromosomal aberration involving TP53BP1 is found in a form of myeloproliferative disorder chronic with eosinophilia. Translocation t(5;15)(q33;q22) with PDGFRB creating a TP53BP1-PDGFRB fusion protein. Variant analysis shows links to various cancers, example: Myomatous Neoplasms
- Sub-cellular location? Nucleus, specifically chromosomes. Localizes to the nucleus in absence of DNA damage. Following DNA damage, recruited to sites of DNA damage, such as double strand breaks (DSBs): recognizes and binds histone H2A monoubiquitinated at 'Lys-15' (H2AK15Ub) and histone H4 dimethylated at 'Lys-20'

- Which PTM? Asymmetrically dimethylated on Arg residues by PRMT1. Methylation is required for DNA binding. Phosphorylations too (for example at basal level in absence of DNA damage.)
- Are there isoforms? 3 isoforms made by alternative splicing . Q12888-1 is the isoform chosen as a canonical sequence. (Q12888-2, Q12888-3 are the other isoforms).
- Orthologs / paralogs?
  Orthologs exist in many species like mouse, frog, zebra fish, yeast, roundworm
  Paralogs: no paralogy available

Ex 3: Screenshot of the WT and Mutant Hemoglobin Superposition (C: beta subunit)

This is what the proteins look like together (no superposition, left):





Normal Hemoglobin **Sickle Cell Hemoglobin**

https://www.discoverymedicine.com/Marcy-C Purnell/files/2020/01/discovery_medicine_no_155_marcy_c_purnell_figure_1.jpg

The disease sickle cell anemia is caused by a mutation in position 6 (single nucleotide substitution: GAG -> GTG) of the beta chain of hemoglobin, point mutation: Glu -> Val. Glutamate is negatively charged and hydrophilic, while valine is nonpolar and

hydrophobic. This creates a sticky surface patch on HbS (mutant): drives polymerization into rigid fibers in the deoxygenated state (aggregation). Global structure slightly changes but mainly conserved (see figure right). The resulting fiber formation makes red blood cells sickle shaped, and by stabilizing the T-state (deoxy conformation), reduces oxygen affinity.